

Saber Ayatollahi, CFA, MSc

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SUMMARY

ALM Director, Consultant candidate with ~7 years developing and validating asset-liability and liability-driven investing models for institutional pension and multi-asset portfolios at Moody's Analytics and Ortec Finance, complemented by insurance-sector regulatory transformation experience (IFRS 17 / IFRS 9) at EY. Holds delegated sign-off authority on valuation, sensitivity, and ALM outputs for multi-asset portfolios of \$5-25bn per engagement; led design of an enterprise multi-scenario cash flow projection engine; and ran VaR/CVaR portfolio optimization, strategic asset allocation, and stochastic scenario analysis for institutional mandates. CFA charterholder with dual MSc and advanced Python.

CORE SKILLS

Asset-Liability Management (ALM) • Liability-Driven Investing (LDI) • Strategic Asset Allocation (SAA) • Portfolio Optimization (VaR / CVaR) • Monte Carlo Simulation & Stochastic Scenario Generation • Cash Flow Projection & Balance-Sheet Modelling • Pension Actuarial Liability Modelling • Duration & Key-Rate Duration (KRD) Analytics • Interest Rate & Liquidity Risk Measurement • Interest Rate, Inflation & Currency Hedging • Stress Testing & Scenario Analysis • IFRS 17 / IFRS 9 Insurance Regulatory Transformation • Python (pandas, NumPy, SciPy) & SQL • Enterprise ALM Platform Onboarding & Client Advisory

EXPERIENCE

Moody's Analytics

Asst. Director - Modelling Services

Toronto, May 2023 - Present

ALM Model Development & Production Processes

- Led design and implementation of an enterprise multi-asset cash flow projection engine supporting base, stress, and reverse-stress scenarios, delivering forward-looking liquidity visibility across ALM horizons.
- Architected configurable time-bucketed liquidity gap analytics (T+1 through multi-year) enabling funding optimization, refinancing risk assessment, and asset allocation decisions.
- Embedded behavioural cash flow assumptions, prepayment logic, and macro stress overlays aligned with regulatory liquidity expectations and executive contingency planning.
- Re-engineered manual spreadsheet workflows into production-quality Python analytics pipelines with logging, validation, and auditability controls, using agentic AI development workflows (Claude Code, Cursor) that cut development cycle time an estimated 30-40% on comparable modules.

Investment Risk Measurement & Analytical Sign-Off

- Held delegated sign-off authority on valuation, sensitivity, and ALM outputs for multi-asset institutional portfolios of \$5-25bn per engagement (~\$50bn+ cumulative), independently reviewing curve construction, spread calibration, and cross-asset interactions before production release.
- Measured interest rate exposure and duration positioning under parallel and non-parallel rate shocks, and reviewed aggregation logic converting security-level exposures into portfolio-level risk metrics feeding downstream ALM and capital processes.
- Validated derivatives pricing outputs across rates, FX, and inflation, confirming sensitivity and scenario consistency at portfolio-level ALM aggregates.
- Served on the model governance committee covering methodology review, documentation and benchmarking standards, and model-performance assessment for client-delivered analytics.

Client Advisory & Communication of Complex ALM Concepts

- Escalated outputs lacking mathematical or economic defensibility and acted as escalation point for client-critical analytical issues, interfacing with clients' Heads of Risk and investment stakeholders.
- Prepared analytical summaries for senior stakeholders on interest rate exposure, scenario impacts, and balance-sheet sensitivities, translating quantitative output into investment-committee-ready narratives.

Client Service Specialist

Toronto, May 2022 - May 2023

Platform Onboarding & Implementation Services

- Led onboarding and implementation for U.S. and Canadian pension funds, asset managers, and consulting firms migrating onto PFaroe DB (risk & ALM analytics) and PFaroe PM (portfolio management).
- Acted as technical liaison between client investment teams and product owners, advising on asset, liability, and ALM configurations, scoping requirements into development requests, and validating model outputs post-deployment.
- Contributed to the successful migration of all assigned client accounts from a legacy Calypso environment to PFaroe PM, reconciling post-deployment output against client requirements.

Ernst & Young (EY)

Senior Consultant - IFRS & Financial Risk Transformation

Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 transformation projects for Canadian and international insurance clients, aligning regulatory reporting requirements with system implementation and financial-reporting frameworks.
- Advised actuarial, IT, finance, and program-management functions on IFRS 17 / IFRS 9 adoption; contributed to governance documentation and regulatory readiness, and developed advisory product offerings and go-to-market collateral.

Ortec Finance Canada

Consultant - Asset-Liability Management & Risk

Toronto, Feb 2019 - Sep 2021

- Delivered ALM studies and liability-driven investment (LDI) analysis for Canadian and international pension funds and institutional investors, analyzing funding ratios, duration mismatch, and long-term balance-sheet risk exposures.
- Performed asset-only and asset-liability (surplus) portfolio optimization on VaR and CVaR using Ortec's GLASS platform, with risk decomposition, contribution-to-risk budgeting, and near-optimal frontier analysis to test robustness of allocation recommendations.
- Built and interpreted stochastic economic scenario generators assessing interest rate risk and funding volatility under base-case and stressed environments.
- Advised clients on strategic asset allocation (SAA) and tactical asset allocation trade-offs through stress testing and sensitivity analysis, and designed stochastic risk-management frameworks for specific client mandates.
- Modelled pension-fund actuarial liabilities alongside all major investment-product types and economic variables, and performed cash flow, liquidity, and hedging analysis across interest rate, inflation, and currency exposures.
- Led onboarding and the ALM study for a three-plan university pension merger on a team of three, developing and validating the model capturing all Funding Policy dynamics and evaluating duration, currency, inflation, and leverage overlay strategies; presented findings to client investment committees on-site.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016